Exercises and Worked Solutions

Part 1: Identification of Statements (True / False)

1. The measurement of area is used to find the amount of space within the boundaries of a plane figure (T)
2. The area of a square is equal to its length. (F)
3. A trapezium is a three-dimensional figure which has a uniform cross-section in the shape of a circle. (F)
4. The total surface area of prisms is the sum of the areas of its faces. (T)
5. The sum of complementary angles is 90°. (T)

Part 2: Worked Solutions

Question 3: Unit Conversions

A) Convert 4 m^2 to cm^2

1 m^2 = 10,000 cm^2

4 × 10,000 = 40,000

Result: 40,000 cm^2

B) Convert 300 m^2 to km^2

1 km^2 = 1,000,000 m^2

300 ÷ 1,000,000 = 0.0003

Result: 0.0003 km^2

Question 5: Surface Area of a Cylinder

Problem: Find the Curved Surface Area (TSA) of a cylinder with radius 5 cm and height 10 cm.

Formula: TSA = $2\pi rh$

Steps:

Radius (r) = 5 cm

Height (h) = 10 cm

$\pi \approx 3.14$

Calculation:

TSA = 2 × 3.14 × 5 × 10

TSA = 314 cm^2

Result: 314 cm^2